

Class 6: R Functions - HOMEWORK

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```
library(bio3d) # necessary library import

analyze_pdb <- function(pdb_ids = c()) { #uses a vector for
  #pdb ids to hold a variable number of pdb ids, generalizable
  #for any number of pdb ids of reasonable length

  for (id in pdb_ids) { # iterate through each PDB ID provided

    pdb <- read.pdb(id) # read the PDB structure from the online
    # database

    chainA <- trim.pdb(pdb, chain = "A", eley = "CA") # limit to
    #c hain A alpha-carbon atoms only

    # secure B-factor values from pdb information
    b_factors <- chainA$atom$b

    # plot B-factors with and include properties of the graph
    # that were used in the original set
    plotb3(b_factors,
           sse = chainA,
           typ = "l",
           ylab = "B-factor",
```

```

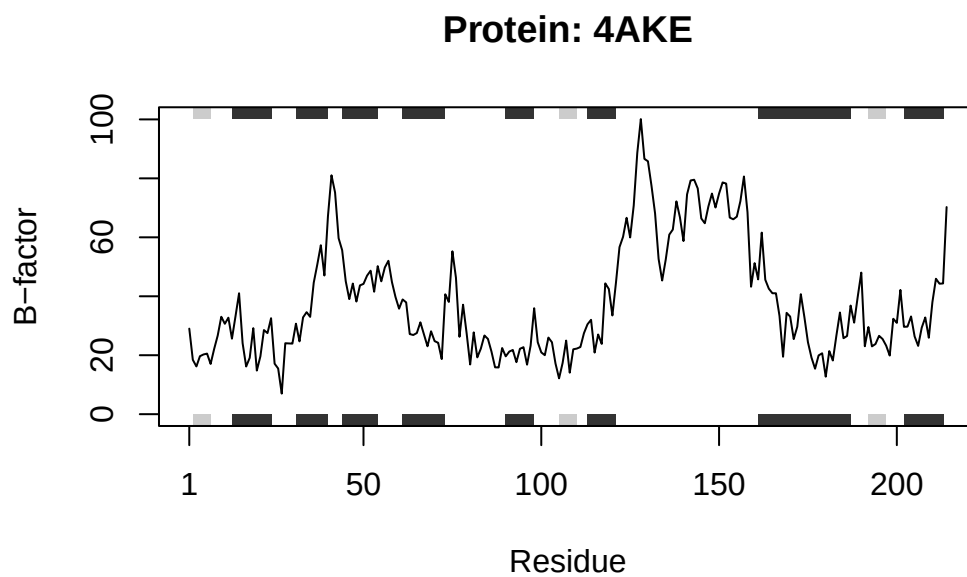
    main = paste("Protein:", id)) #this last line adds
# an annotation that states which protein this graph is for
# and what its pdb ID is

#the graph is realistically the only function output
#There is no indication of a necessary value or values to be
#returned
}
}

# Example usage:
analyze_pdb(c("4AKE", "1AKE", "1E4Y"))

```

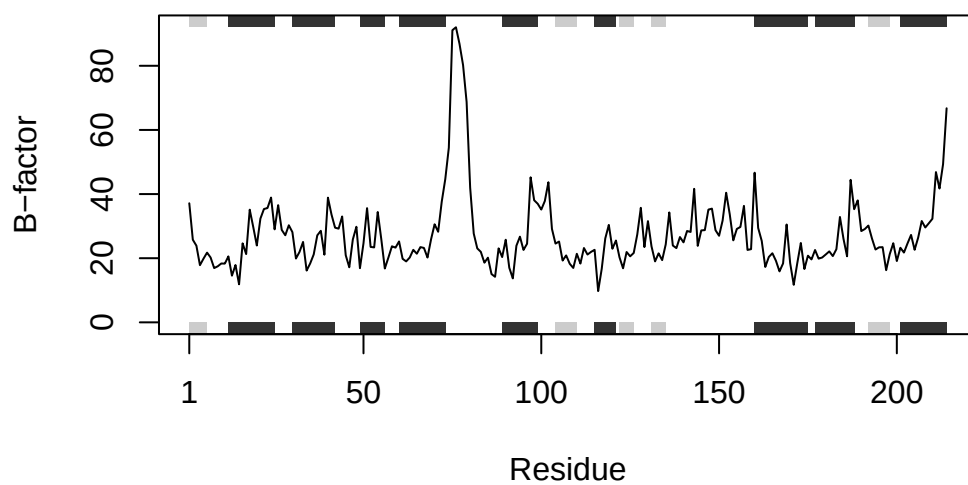
Note: Accessing on-line PDB file



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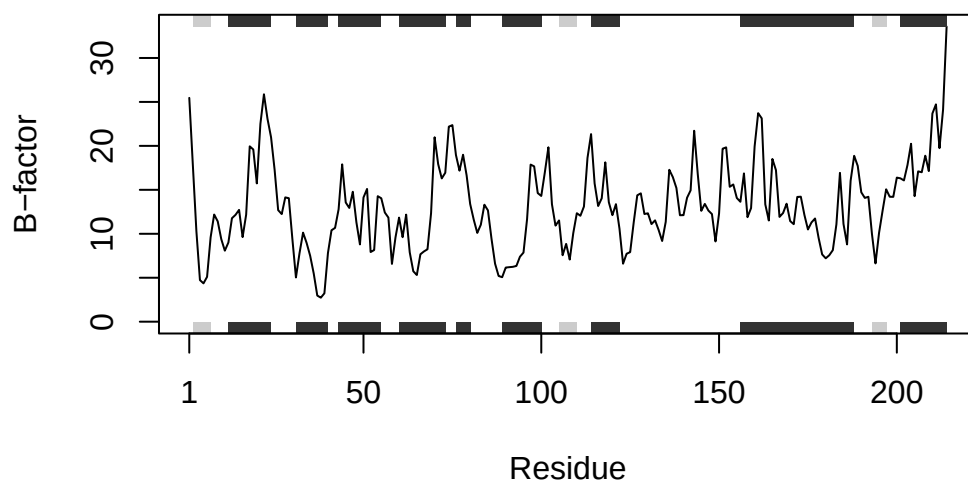
PDB has ALT records, taking A only, rm.alt=TRUE

Protein: 1AKE



Note: Accessing on-line PDB file

Protein: 1E4Y



```
#example output would include 3 graphs:  
#each graph has a title protein: and its id:  
#Protein 4AKE, and its relevant graph with B-factor on the  
#y axis and residue on the x  
#Protein 1AKE, and its relevant graph with B-factor on the  
#y axis and residue on the x  
#Protein 1E4Y, and its relevant graph with B-factor on the  
#y axis and residue on the x
```